Homework Solutions

Course Math 1A Worksheet 4 · **Student** Songpengfei · **Date** 2026-10-09 · **Solved** 9/10 (90.0%)

Problem Q1

Suppose f is a function and L and a are real numbers.

1. See sub-problems for individual limit computations.

   (a) $\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

   (b) $\textbf{See graph}$

   (c) $\textbf{See graph}$

   (d) $\textbf{See graph}$

> **Answer** $\textbf{See sub-problems}$

`sympy_limit` calculus_limits

Problem Q2

Is ∞ a number? What does $\lim_{x \to a} f(x) = \infty$ mean? Does $\lim_{x \to a} f(x) = \infty$ mean that the limit exists?

1. ∞ is \textbf{not} a real number.
2. $\lim_{x \to a} f(x) = \infty$ means that $f(x)$ grows without bound as $x \to a$.
3. The limit does \textbf{not exist} in the usual sense; we say the limit ``equals infinity'' as a shorthand for divergence.

> **Answer** $\infty \textbf{ is not a number. The limit does not exist.}$

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Problem Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

1. \textbf{Squeeze Theorem:} If $g(x) \leq f(x) \leq h(x)$ for all x near a (except possibly at a), and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$, then $\lim_{x \to a} f(x) = L$.
2. Intuitively: if f is ``squeezed'' between two functions that both converge to L, then f must also converge to L.

> **Answer** $\text{If } g(x) \leq f(x) \leq h(x) \text{ and } \lim g = \lim h = L, \text{ then } \lim f = L.$

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Problem Q4

What do we mean by $\lim_{x \to a^-} f(x) = L$? What is meant by $\lim_{x \to a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$. Find an algebraic expression for such a function.

1. $\lim_{x \to a^-} f(x) = L$: the \textbf{left-hand limit}. As x approaches a from values \emph{less than} a, $f(x)$ approaches L.
2. $\lim_{x \to a^+} f(x) = L$: the \textbf{right-hand limit}. As x approaches a from values \emph{greater than} a, $f(x)$ approaches L.
3. The two-sided limit $\lim_{x \to a} f(x) = L$ exists if and only if both one-sided limits exist and are equal.
4. Example: $f(x) = \begin{cases} x & x < 0 \\ x + 1 & x \geq 0 \end{cases}$ at $a = 0$:
   $\lim_{x \to 0^-} f(x) = 0 \neq 1 = \lim_{x \to 0^+} f(x)$.

> **Answer** $\lim_{x \to a^-} f(x) : \text{from below}; \quad \lim_{x \to a^+} f(x) : \text{from above}$

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Problem P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x\to a}[f(x)+g(x)]$ exists even though neither $\lim_{x\to a} f(x)$ nor $\lim_{x\to a} g(x)$ exist.

1. No, the limit of a sum is \emph{not} always the sum of the limits.
2. The \textbf{Sum Law} requires both individual limits to exist: $\lim[f(x)+g(x)] = \lim f(x) + \lim g(x)$ \emph{if both limits exist}.
3. \textbf{Counterexample:} Let $f(x) = \frac{1}{x}$ and $g(x) = -\frac{1}{x}$. Neither $\lim_{x\to 0} f(x)$ nor $\lim_{x\to 0} g(x)$ exists, but $\lim_{x\to 0}[f(x)+g(x)] = \lim_{x\to 0} 0 = 0$.

Answer $\text{No. } f(x) = 1/x,\ g(x) = -1/x : \lim[f+g] = 0 \text{ but } \lim f, \lim g \text{ DNE}$

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Problem P2

(a) Find $\lim_{x\to 0^+}(\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x\to 0^-}(\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x\to 0}(\frac{1}{x} - \frac{1}{|x|})$, if they exist.

1. $\lim_{x\to 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$
2. $\lim_{x\to 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$
3. $\lim_{x\to 0} -\frac{1}{|x|} + \frac{1}{x} = 0$

(b) 0

(c) DNE

Answer 0

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Problem P3

For what kinds of functions can one evaluate the limit $\lim_{x \to a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \to a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

1. \textbf{Continuous functions} allow direct substitution: $\lim_{x \to a} f(x) = f(a)$.
2. This includes polynomials, rational functions (at non-zero denominator points), trigonometric, exponential, and logarithmic functions at points in their domain.
3. Example where limit exists but substitution fails: $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$. Direct substitution gives $\frac{0}{0}$, but $\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} (x + 1) = 2$.

> **Answer** $\textbf{Continuous functions: } \lim_{x \to a} f(x) = f(a)$

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Problem AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

1. See sub-problems for individual limit computations.

(a) -20

(b) 10

(c) 0

> **Answer** $\textbf{See sub-problems}$

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Problem AP2

Using a calculator, try to guess $\lim_{n\to\infty} n\sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

1. $\lim_{x\to\infty} n\sin\left(\frac{\pi}{n}\right) = \pi$

> **Answer** π

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Problem AP3

Draw a circle of radius 1. Inscribe a regular n-gon inside it.

> ⚠️ **未解出**：确定性求解器未覆盖，且本次运行禁用了 LLM（--no-llm）